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NOTES ON POST PALEOZOIC OSTRACODA

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I. STRATIGRAPHIC AND PALEOECOLOGIC USEFULNESS OF POST-PALEOZOIC OSTRACODA

The value of Paleozoic Ostracoda in economic paleontological studies has long been recognized, but it was generally believed that the Mesozoic and Cenozoic representatives of this group of microfossils were of little stratigraphic interest. On the North American continent, both within and outside our Company, they have so far received only very scant attention, although, like almost everywhere else, they occur rather regularly in most sedimentary rocks and in some parts of the stratigraphic section they not rarely outnumber the Foraminifera.

The general belief that Mesozoic and Cenozoic Ostracoda constitute a less valuable tool for stratigraphy than the more commonly used Foraminifera finds its origin in the fact that most earlier workers were insufficiently acquainted with the structural details of their calcareous valves. They grouped many related or superficially similar forms, of different geological age, under one specific name. This led to the supposition that most Post-Paleozoic Ostracoda were long-ranging forms and therefore of little or no value as stratigraphic markers. However, subsequent detailed studies revealed much additional information on the morphology of the valves and in the past few years it has been proved conclusively in many relevant publications that the Post-Paleozoic Ostracoda are certainly of great stratigraphic and paleo-ecologic value. In the Royal Dutch/Shell Group the importance of the Ostracoda was realized quite some years ago, and a special Ostracoda Research Section was established in the Central Paleo. Lab. in the Hague in 1955. At the present time they are used to great advantage by our Group paleontologists in many different parts of the world, and it is hoped that this short presentation will result in their getting more attention from our paleontologists here in North America as well.

Ostracoda indeed possess many of the qualifications required of a practical micropaleontological tool. These may be briefly mentioned as follows:

1. They are found in every type of aquatic environment

Besides occurring in all types of marine environments, Ostracoda are also commonly encountered in brackish and freshwater deposits. Unlike the Foraminifera, therefore, Ostracoda often form an excellent group of fossils for the correlation of marine sediments with brackish to continental deposits. Although these various biotopes are not generally inhabited by the same Ostracod taxa or associations, reliable correlations are nevertheless usually possible as a result of the fact that the freshwater, brackish and marine Ostracod taxa and assemblages generally show an appreciable overlap. For paleo-ecological interpretations, this overlapping of the thanatotopes must, of course, be regarded as a handicap, rather than an advantage, but from a stratigraphic point of view, the synchronous co-deposition after death of Ostracoda which lived in such different environments is very convenient.

2. The range of variation within a species is small.

The calcareous valves of Ostracoda, when not too badly preserved, exhibit a wealth of structural details, allowing a precise delimitation of the taxonomic units. The range of variation of various morphological features of the valves in synchronous specimens of the large majority of species would appear to be very small. This applies also to the more readily preserved and observable features like size, shape and external ornamentation, thus rendering correct specific determinations easier and less dependent on large numbers than, for example, in the Foraminifera. Even fragments of valves may often be correctly identified specifically.

3. Their vertical range is generally short

Ostracoda are highly organized animals. They form part of the very specialized phylum Arthropoda, and belong to the same class, Crustacea, as, for example, the Trilobita, whose great stratigraphic value is doubted by no one. Compared to the uni-cellular Foraminifera, the Ostracoda generally seem to have passed through a more rapid phylogenetic development. The close contact between body and calcareous carapace would appear to have caused the calcareous valves, too, to reveal in many cases a series of rapid, well-marked morphogenetic changes. These may be considered to be a function of geologic time, and may be used for detailed synchronous correlations of profiles in which they are found. Thus, the genera, and especially the species, if properly defined, generally exhibit a restricted vertical range, and slight modifications within the species at different stratigraphic levels may often enable a chronological subdivision to be made in finer detail than is normally achieved with the aid of Foraminifera.

Some main morphogenetic trends, pointing to a parallel phylogenetic development, may be observed in various structures of the calcareous valves of the different groups of Ostracoda during Post-Palaeozoic time. They are mentioned here because they sometimes allow a quick inference of the approximate geological age of an assemblage before any generic or specific determinations are made.

4. Their horizontal distribution is extensive

Practically all the Ostracoda found fossil lived a benthonic life. Pelagic representatives useful for world-wide synchronous correlations, such as are found amongst the Albian-Recent Foraminifera, are not known. Throughout the geological column, however, from the Jurassic to Recent, a number of typical genera of rather restricted vertical range occur; often of inter-regional or world-wide horizontal distribution. Recognition of such approximate time-markers is relatively easy and hardly requires specialized training. Ostracod species are generally less valuable for long-distance correlations. Many species lived in more or less restricted areas and are therefore only of regional stratigraphic value.

5. They are not at all rare

It is often argued that Ostracoda are less commonly encountered in samples than are the Foraminifera. This is undoubtedly true as far as samples of marine origin are concerned. However, in the case of freshwater sediments, and generally also brackish water sediments, the reverse is true. In marine samples, Ostracoda generally occur in sufficient quantities to be of use, and after the Foraminifera they constitute on the whole the second most commonly occurring group of fossils. The percentage of marine samples without Ostracoda is much lower than is generally believed and probably does not exceed 25 percent when samples are examined somewhat more extensively.

Despite the obvious advantages of using Ostracoda in our stratigraphic studies, it is a well known fact, though a sad one, that perhaps 90 percent of all micropaleontologists in practical work devote their attention exclusively to the Foraminifera and disregard the Ostracoda. We feel very strongly that no group of microfossils should be disregarded, whether or not a micropaleontological subdivision of the section on one group of fossils has already been established. Many of those not familiar with Ostracoda believe that an extensive training and background is required in order to make use of these fossils. This is an erroneous conception. Any micropaleontologist with a trained eye for fine morphological details in, for example, the Foraminifera, can apply this skill equally well to the fossil Ostracoda.

II. STRUCTURAL ELEMENTS IN OSTRACODS

1. General description

The Ostracoda form a sharply defined order of the class Crustacea, Phylum Arthropoda (see Classification, p. 36).

The most conspicuous feature of the Ostracods is a bi-valved carapace into which the soft body may be wholly withdrawn. This gives the Ostracods an outward resemblance to the bivalved molluscs, hence the names "Mussel shrimps", "Muschelkrebse" (German) or "Schaalkreeftjes" (Dutch).

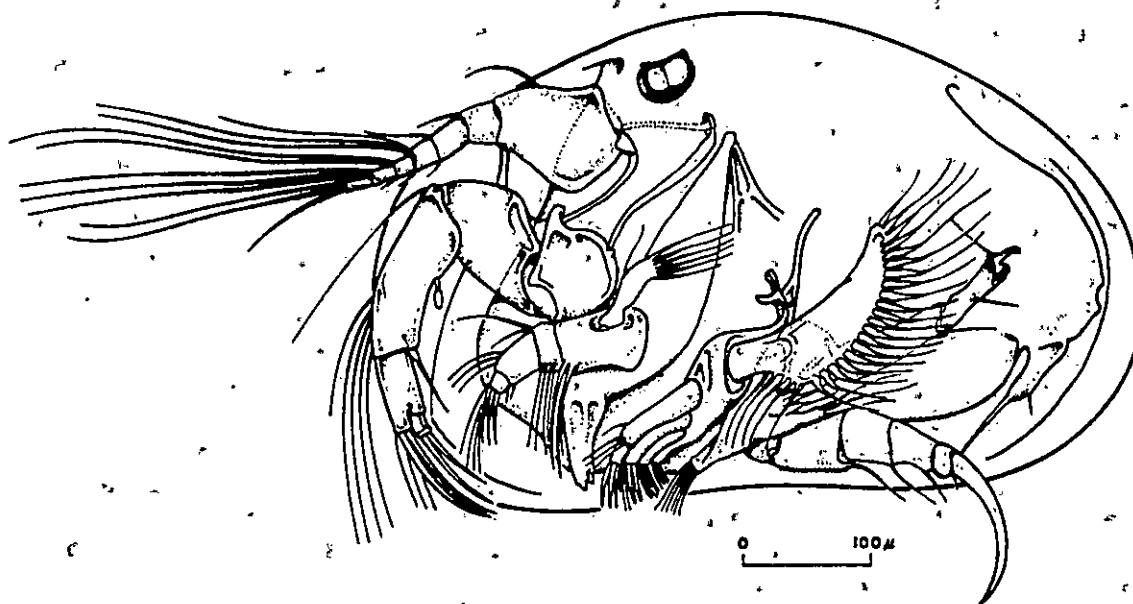
The carapace, however, is no mere externally secreted shield - as it is in the Lamellibranchs - but forms an integral part of the chitinous covering of the epidermis.

The body of the Ostracods conforms to the generally sub-ovate shape of this enclosing shell. Apart from the appendages, the various structures of the animal do not show close affinities with those of the higher Crustacea.

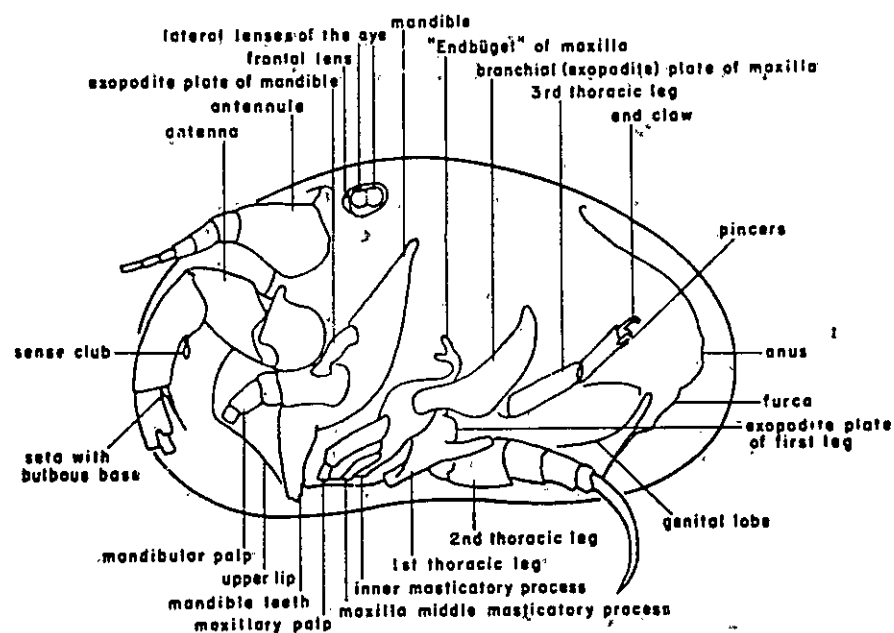
The compact short body can best be seen after removal of one of the valves (see fig. 1). There is no true segmentation as often found in the Arthropoda, although there is a slight constriction of the body near the middle, marking the boundary between its two main portions:

the cephalon (head) and
the thorax.

The abdomen is rudimentary and fused with the thorax.



Cypridopsis vidua (O. F. MUELLER)



Labelled diagram of the above figure

Fig. 1: Lateral view of an Ostracod in its shell (left valve removed) showing position of appendages.
(after KESSLING, 1951)

There are usually seven pairs of appendages:

1st Antennae (or Antennules)	)	
2nd Antennae	)	attached to
Mandibles	)	the cephalon
Maxillae	)	
1st Thoracic legs	)	
2nd Thoracic legs	)	attached to
3rd Thoracic legs	)	the thorax

The body occupies the middle two-thirds of the length of the carapace. It is suspended from the dorsal region as an elongate chitinous pouch. It is highest and rather blunt at the anterior end, tapers slightly throughout its length, terminating abruptly at the posterior end where a furca is attached. This furca is usually not regarded as being an appendage.

The outer wall of the body, commonly termed the exoskeleton, consists of an outer chitinous layer and an inner epidermal layer. In life, the actual body of the Ostracod is enclosed in two lateral "flaps" or "duplicatures" of this body wall. These originate on each side of the body near the boundary between cephalon and thorax (fig. 2). Each duplicature consists of an "outer lamella" and an "inner lamella".* The space between outer and inner lamella must be regarded as a lateral extension of the actual Ostracod body, and it often contains various vital organs (livers, ovaries/testes, etc.). The epidermal layer of the outer lamella secretes a hard, more or less heavy calcareous layer over all of its surface, which in turn is coated with a thin chitinous layer. The peripheral part of the inner lamella is usually more or less calcified as well.

Duplicature, calcareous layer and chitin outer coating together form the ostracod valve.* The left and right valve together are known as the carapace. As a rule they are not symmetrical; the left one being normally somewhat larger and overlapping the right one. The outer surface of the valves may be ornamented or smooth.

Unlike the shell of the Lamellibranchs, which grows continuously, the calcareous and chitinous layer of the Ostracods is periodically shed and replaced by a new - larger - one. This process is known as "moulting".

The two valves are hinged together at the dorsal margin by a ligament. The hinge-margin may be either simple or complicated by ridges and grooves and/or teeth and sockets. The hinge elements, if present, in the one valve are usually complementary to those in the other valve. Throughout the rest of the circumference the valves are not attached, so that the animal has free access to the water when the valves are open.

* When referring to the living animals the biologists use the terms inner lamella, outer lamella and valve for hard and soft parts together. In dealing with fossils these terms are usually applied to the hard calcareous parts only.

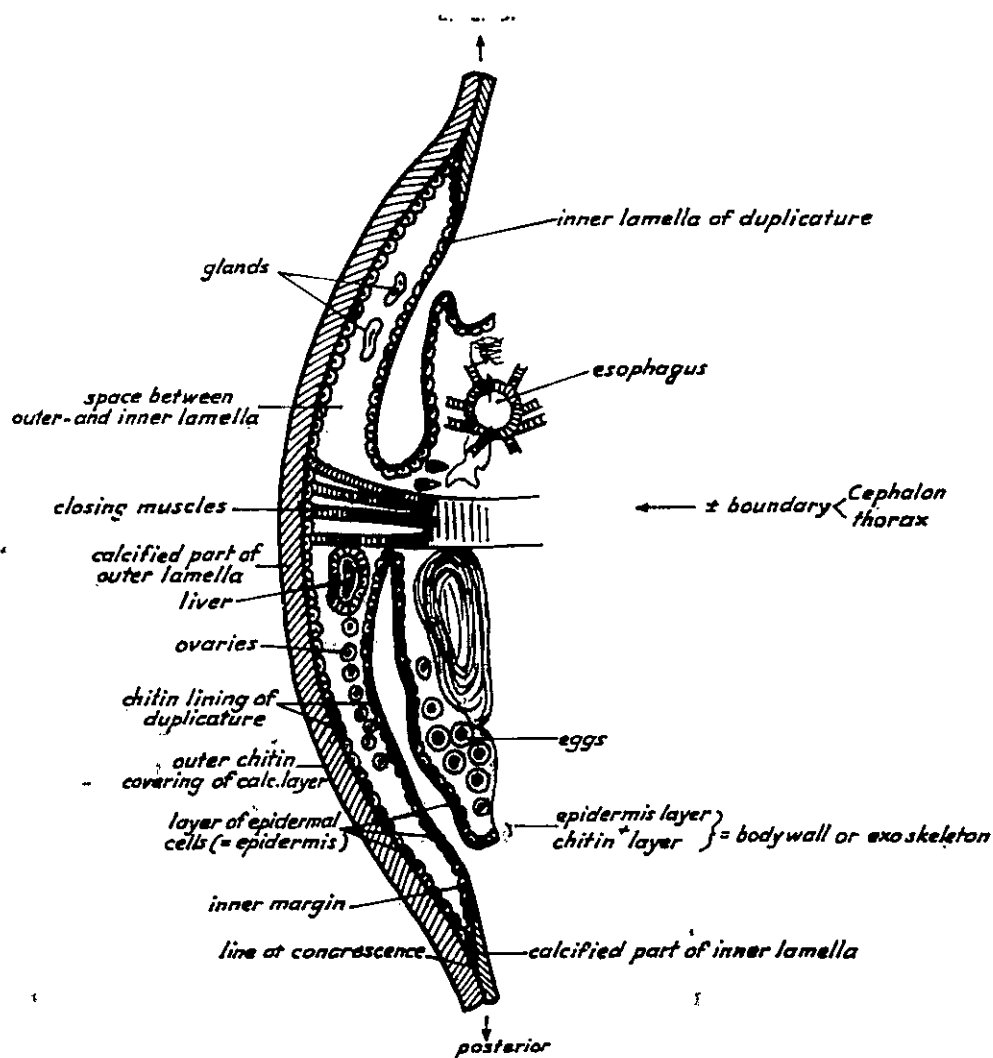


Fig. 2 Longitudinal median section through the left half of a recent ostracod showing nature of duplicature and distribution of calcareous parts.

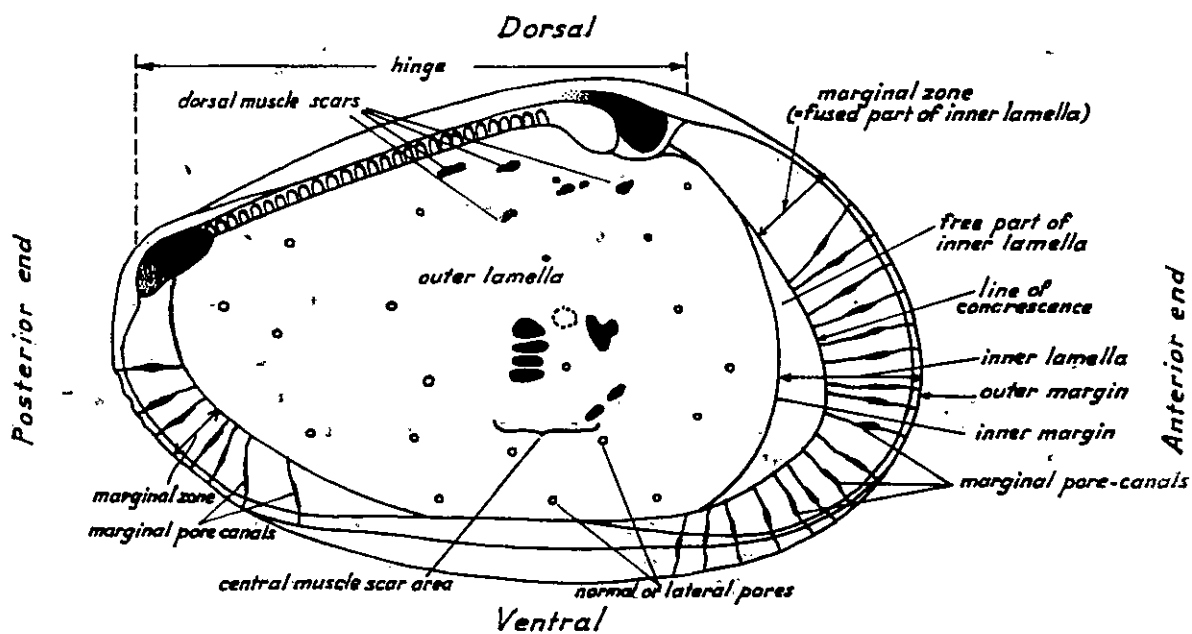


Fig. 3 Left valve of an ostracod, seen from the inside, showing main structural elements of the calcareous parts.

The carapace is opened and closed and certain appendages (mandibles, antennae) are operated by muscles attached to the inside of the valves. The points of attachment of these muscles are often visible on the calcareous shell and are known as "muscle scars" (fig. 3).

The calcified part of the inner lamella may be wholly or partly fused with the calcareous layer of the outer lamella. The fused area is known as the "marginal zone"; it is transversed by a varying number of marginal pore canals, running from the inner space of the duplicatures to the outer world. Additional pores, piercing the lateral surface of the outer lamella, are known as "normal pores" or "lateral pores".

The sexes in Ostracods are always separated. The males, however, are often missing, in which case reproduction takes place parthenogenetically. Sexual dimorphism is often also apparent in the calcareous valves.

2. Criteria for determining ostracod genera and species (in order of importance)

Genera

1. Shape (outline, presence of certain lateral projections and depressions)
2. Type of ornamentation
3. Hinge type
4. Presence or absence of eye spots
5. Finer details of the central muscle scar area
6. Course of the line of concrescence
7. Number, type and general distribution of the marginal pore canals and lateral pores
8. Overlap
9. Width and structure of the marginal zone
10. Presence on the valves of opaque spots

Species

1. Details of ornamentation
2. Position and number of marginal pores
3. Variations of shape
4. Development of selvage; lists, etc.
5. Size
6. Finer details of the hinge
7. Distribution and number of lateral pores
8. Distribution and pattern of opaque areas in the shell
9. Presence or absence - and number - of antero-and/or postero-marginal denticulations or spines
10. Overlap

N. B. Specific, and to a lesser degree generic determinations are made difficult in many cases by

1. sexual dimorphism
2. ontogenetic polymorphism.

3. Shape

The shape of the Ostracod carapace may be influenced and altered by:

1. the presence of lateral extensions or indentations of the valves
2. ornamentation
3. the process of moulting
4. sexual dimorphism
5. environment

V. d. BOLD (1946) has coined the term "aerodynamic shape" for Ostracods. This term is self-explanatory*, and indicates that the Ostracods, in order to better overcome the friction and resistance of the surrounding water, have succeeded generally in "streamlining" their carapace in such a way that pointed ends are found at the rear only, and that all lateral projecting elements are pointing backwards.

For easier reference the Ostracod valve may be subdivided into a number of areas or regions designated as shown in fig. 4.

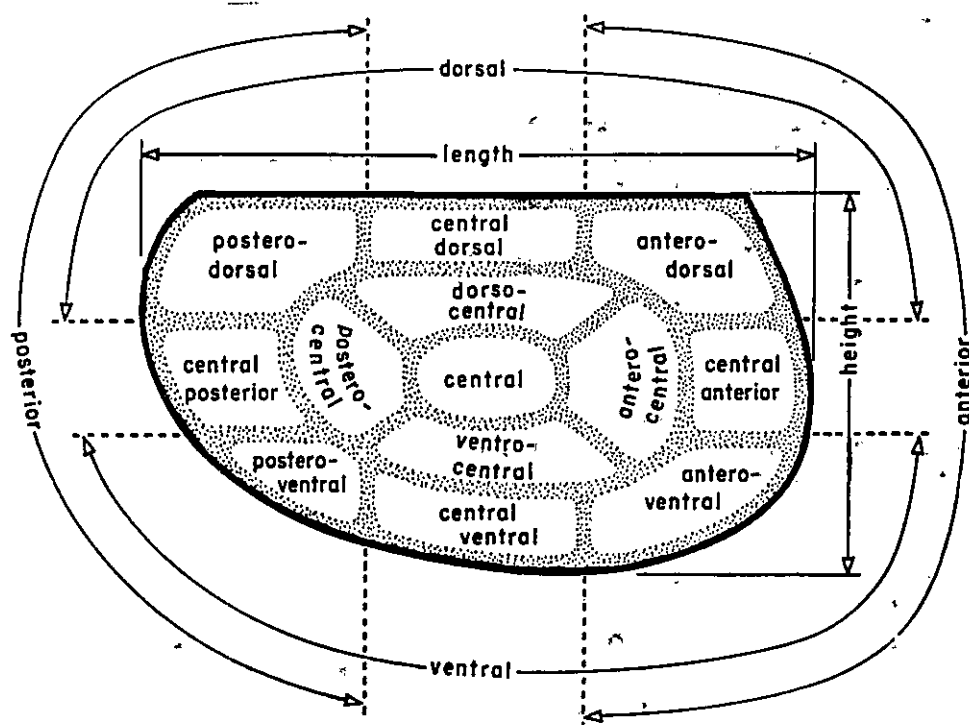


Fig. 4 : Terminology of the lateral surface of an Ostracod valve (after KESSLING, 1951).

* Although "hydrodynamic" would be a more fitting term.

The outline of the Ostracod carapace (or of one valve) is here defined as the profile of the carapace (or valve) when seen in lateral view. It embraces all features of the carapace (or valve) such as alae, subventral swellings, prominent ornamentation, etc. It is necessary, therefore, to distinguish between outline and outer margin. The latter is the actual outer periphery of a valve. The outer margin, in lateral view from the outside, may be partly invisible or masked by strong lateral projections or ornamentation, and its course may therefore be best observed in lateral view from the inside. Often, however, outline and outer margin coincide in lateral view.

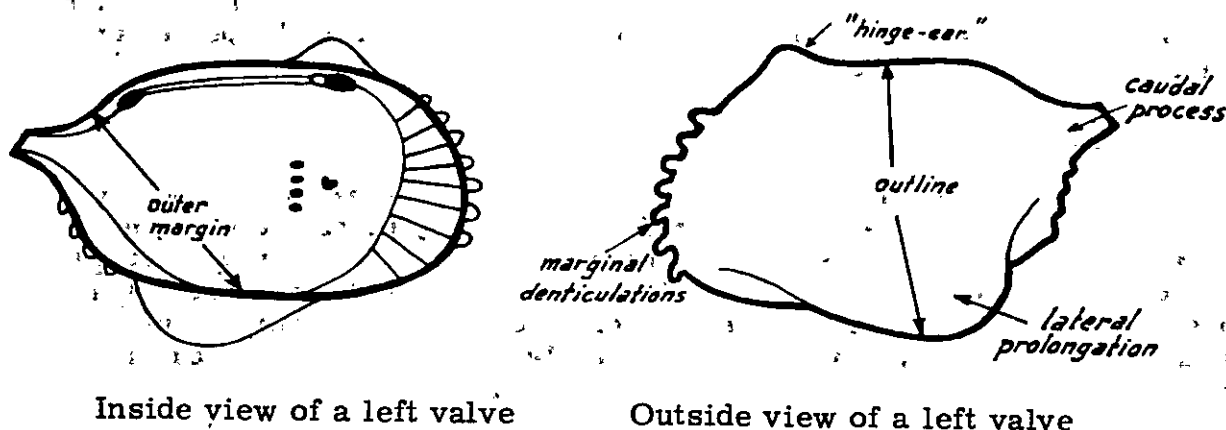


Fig. 5: Showing difference in outline and outer margin, and certain prominent features of the outline. (composite drawing)

The following marginal features may constitute part of the outline of the Ostracod carapace or of single valves:

Marginal denticulations ("Randzähne")

In many species the outer margins of the valves are provided with a number of characteristic, toothlike prolongations of the shell wall. Their number, shape and dimensions may vary. They are usually found at the anterior margin (especially antero-ventrally) and/or at the posterior (especially the postero-ventral) margin. More rarely they occur all along the margin of the valves, or in one valve only. Sometimes the marginal teeth may be interconnected by a thin calcareous lamellar structure. In most cases they originated probably by a subdivision of the flange into separate elements. Normally they are not considered as part of the lateral surface ornamentation.

Caudal process

In certain Ostracods the posterior end of the carapace terminates with a drawn-out, nozzle-like prolongation, whereby, when the carapace is closed, a small opening usually remains for the circulation of the water. This process is referred to as a caudal process. It may be situated either posterodorsally, posteroventrally or posterocentrally.

Posteroventral spine

A conspicuous spine-like calcareous appendix is found sometimes attached to the posteroventral angulation of the smaller valve (certain spp. of Cyprideis, Paracyprideis etc.). More rarely there are two of such posteroventral spines either both on one valve, or one on each valve.

"Hinge ears"

A strengthening of the outer (dorsal) margin of the terminal sockets in the larger valve (by additional secretion of calcareous material) may result in pronounced arched extensions of the dorsal margin of the larger valve. These are often referred to as "hinge ears". They are more common in the anterodorsal region, but occur sometimes also above the posterior hinge socket, or above both terminal sockets.

Eye tubercles

In certain cases the outline of the carapace may be profoundly changed by the development of extraordinarily large, upward pointing eye tubercles. Such is the case, for example, in the genus Exophthalmocythere, an exclusively fossil form from the Jurassic and Lower Cretaceous. Although this feature has been used as the diagnostic criterion for this genus, it is possible and likely that it may eventually also be found to occur in other, genetically not related forms.

Anteroventral beak - Rostrum

In some genera of the extinct Cyprid subfamily Cyprideinae the anteroventral margin exhibits a marked, abrupt indentation on each valve, while the surrounding lateral area of the valves is marked by shallow vertical notches. Together these features are known as an "anteroventral beak". It constitutes a very characteristic part of the outline and distinguishes these genera (Cypridea and allied forms) immediately from all other Podocopa.

A somewhat similarly shaped, but essentially very different feature is present in practically all Myodocopa. Here we find an anterior opening in the carapace, together with incision-like continuations of that opening in the lateral surface of each valve. It is responsible for a peculiarly shaped anteroventral outline, useful for distinguishing the calcareous parts of the Myodocopa from all other suborders, and is called a "rostrum" (see fig. 6).

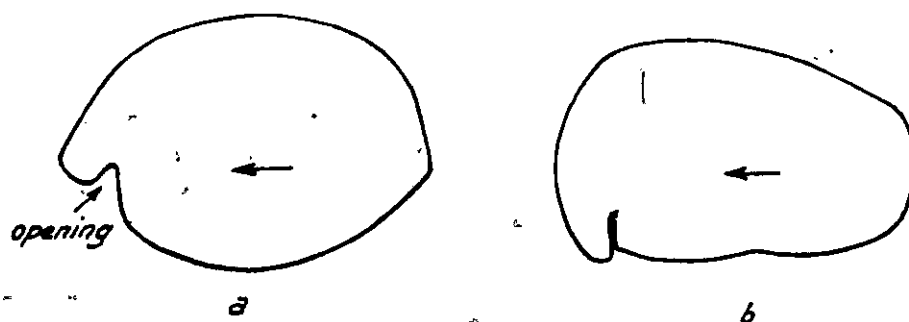


Fig. 6 : a: outline of Cypridina showing antero-ventral "rostrum"
b: outline of Cypridea showing antero-ventral "beak"

Size of the Ostracod carapace

The Ostracod carapace is relatively small. Due to its small dimensions it often occurs in large numbers in borehole cuttings and cores, and is therefore an important aid in micropalaeontological studies for chronological/ecological purposes. Ostracods would seem to be the most highly organized microfossil organisms encountered in such samples, being equivalent in rank, for example, to the much larger Trilobita.

Except for the Myodocopa, the length of most Post-Palaeozoic Ostracods varies between 0.4 and 2 mm; smaller forms are less common and larger species may even be said to be rare. The latter practically all belong to the freshwater Cyprididae. The recent Myodocopa comprise quite a few species and genera with a size exceeding 2 mm, but representatives of this suborder are hardly ever found in the fossil state. One of the largest living Ostracod species would seem to be Gigantocypris agassizii G. W. MULLER (Myodocopa) from the west-coast of Central America, of which the length is approximately 21 mm*. It is difficult to say which is the smallest Ostracod; Nannokliella dictyoconcha SCHÄFER, for example, attains only 0.14 mm, but it is likely that smaller species do exist.

Overlap

As a rule one valve is larger than the other. In most of the Podocopa the largest valve is the left one, which overlaps the smaller right one along the whole periphery, more rarely only dorsally and ventrally so. In the Cytherellidae the right valve is characteristically the largest, as is also the case in some genera of the Cyprididae (Macrocypris, Argilloecia, etc.). The overlap seems to be a constant feature for a species (except for certain species where overlap and hinge structure are reversed. In quite a few genera some species have the left valve overlapping the right one, while other species show the reverse (Cypridea, Darwinula, Cypridopsis, etc.).

* In Palaeozoic Ostracods even much larger forms are not at all rare. Leperditia titanica SCOTT, for example, measures 58 mm!

Taxonomic importance of shape, outline and size

Shape or outline, or parts thereof, are often characteristically constant in species, genera, and even in systematic categories of higher rank. Good examples of this are plentiful, such as the "rostrum" of the Myodocopa and the typical subcircular outline and lenticular shape of the Cladocopa. The very characteristic shape of Bairdia is found in many other genera of the family Bairdiidae as well, and the special term "bairdioid shape" has been created for this particular type of outline.

In the Cyprididae and Cytheridae, genera of a same subfamily often show a marked similarity in outline features. All Cytherurinae, for example, exhibit a more or less pronounced caudal process; an anteroventral beak is found only in the Cyprideinae; the genera belonging to the Bythocytherinae are all thin-shelled and most have a typical straight hinge line, etc., etc.

At generic level the outline is of outstanding importance as a diagnostic feature. Recent species, assigned to the same genus on account of their soft parts, in nearly all cases show such close similarity of outline that this feature alone is often sufficient to recognize the genus. The taxonomic importance of the outline as a generic diagnostic feature (probably the most reliable and most constant one) cannot be overstressed, although in Ostracod literature this point has not always been given the emphasis it deserves. In generically related species the outline is not, of course, always exactly similar in all details, but it generally offers so many corresponding points that in practical work it is a most useful tool for generic determinations. The presence of such features as hinge ears, an anteroventral beak, large eye tubercles, and the presence (and especially the position!) of a caudal process or otherwise pointed posterior end are constant within a genus. Species with a conspicuously long straight hinge margin, for instance, in all probability are not generically related to others with a curved dorsal margin, and very likely do not even belong to the same subfamily.

As is usual, this rule has its exceptions. Certain genera, such as Candona contain species which show a notable divergence of outline as compared to others. On the whole, however, such exceptions are not very numerous, and they detract only little from the value of shape as a taxonomic character.

• Cases of homeomorphism (i.e. similar shape and type of ornamentation in different genera) are also known in Ostracods. It occurs more often in the smooth featureless freshwater Cyprididae than in the marine ornamented Cytheridae. In the latter family a good example of homeomorphy is found in the genera Schizocythere and Cnestocythere (see TRIEBEL, 1950, and fig. 7). Generic differences in such homeomorph genera are found mainly in the structure of the hinge.



Fig. 7: Example of homeomorphy in Ostracods.

1. Cnestocythere truncata (REUSS)
2. Schizocythere hollandica TRIEBEL

Variations in outline and shape between adult specimens of one species (apart from those due to sexual dimorphism) are as a rule, extremely slight, and practically negligible. Somewhat more variable in certain species is the individual size. Here too, however, variation seems to be smaller than is generally believed. In most species it probably does not exceed 5%. "Dwarfed" faunas (as known in foraminifera) resulting from unfavorable living conditions; have not been observed so far in Ostracods*. Of the large variations in size mentioned in literature, many are probably due to heterogeneous species determinations, or to assemblages including both instars and adults, or representatives of the two different sexes.

At generic level, size is a less valuable taxonomic aid than outline, but nevertheless not altogether negligible. Many genera contain only relatively small, or only large species. Variation in size of congeneric species, at a rough estimate, may range up to 100%; it is believed, however, practically never to reach such abnormal proportions as 250% or even more. On suprageneric level size does not seem to be of any practical taxonomic use.

Relation between shape, size and mode of life

Certain authors (REMANE, 1933, 1940, ELOFSON, 1941) have carried out detailed studies on the possible relationship between the shape and size of Ostracods and the nature of the substratum on which they live. These studies resulted in a number of rather detailed conclusions. However, their observations were confined mainly to a comparatively small group of marine benthonic Ostracods from the Baltic Sea region, and very few similar observations have been put on record from other areas such as, for example, the tropical seas.

* An exception must be made for certain marine species which occur in the arctic waters as well as in waters of the temperate zone. The arctic representatives were found to be considerably larger than the latter; in some cases about twice as big. This phenomenon is also known in the mollusca. It is difficult to decide which is the true habitat of such forms; in other words: whether the temperate zone forms must be regarded as "dwarfed" forms, or vice versa.

On the whole, too few data are as yet available about this subject, and at the present moment it seems unwarranted to apply indiscriminately all such conclusions about the shape, size / mode of life relationship in Ostracoda to the whole order.

Nevertheless, there is little doubt that a relationship of this kind does exist to a certain extent, and environmental factors have certainly influenced the shape of Ostracod carapaces. Their generally "aerodynamic shape" is, of course, not accidental. The wing-like ventro-lateral extensions of the valves found in certain genera (Pterygocythereis, Cytheropteron, etc.) and/or the pronounced flattening of the ventral surface of the shell (Brachycythere, Paracytheridea, etc.) provide additional support for such forms while crawling over the surface of a soft muddy bottom. The flat ventral side of Notodromas is perfectly suited for hanging upside-down at the surface of the water.

Apart from such obvious adaptations to environment, however, it is at present not possible to provide a reliable overall set of rules with which conclusions of practical value about environment, substratum, etc. might be deduced from the shape and/or size of fossil Ostracods.

4. Ornamentation

The lateral outer surface of the Ostracod valves may be either smooth or ornamented. Ornamentation is here defined as the total of all local elevations and depressions (thickenings or thinnings) of the calcareous valves at their outer lateral surface, that are not reflected on the inner surface. Sub-central tubercles, sulci, lateral extensions, and hollow tubercles caused by environmental factors (e.g. as often found on the brackish-water Cytherideinae), therefore, are not included here under the term ornamentation).

As a rule, the ornamentation on right and left valve is the same, but certain exceptions are known to occur (e.g. in the males of Carinocythereis, Occultocythereis and Hermanites). Different ornamentation between males and females of a species is a less rare, but even so not common feature. Differences in ornamentation between the various instars and the adult are not unusual.

Ornamentation of the valves of Ostracods may consist of punctations, pits, spines, massive protuberances (tubercles or knobs), striae, ridges or costae (transversal, longitudinal or concentric), reticulation, etc. or combinations thereof*. No special terminology is necessary for each particular part or type of ornamentation. It is worth mentioning, however, a few self-explanatory terms for parts of one rather frequently occurring type of ornament, viz. that formed by the presence of three prominent longitudinal ribs. These are known as dorsal ridge, median ridge and ventral ridge. Often this type of ornamentation is accompanied by a conspicuous anteromarginal ridge, and/or a subcentral tubercle (see fig. 8).

* Some examples of ornamentation are shown in Fig. 11 (orientation) i.e. pitted (B), reticulate (G), with concentric ridges (F), with three longitudinal ridges (C), etc.

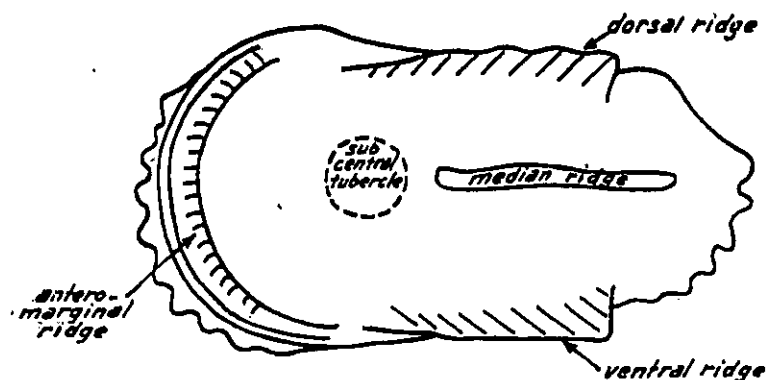


Fig. 8: Position and terminology of some common ornamental ridges.

A peculiar type of ornamentation sometimes develops in certain reticulate genera as, for example, Spongicythere, Cytheralison, Platycythereis, etc. The small ridges on the outer lateral surface of the valve, which form the reticulation, begin to extend laterally at their tops. The result is a peculiar "sponge-like" type of superstructure, consisting of a sort of porous second calcareous layer supported and separated from the outer surface of the actual outer lamella by the vertical ridges of the reticulation. (fig. 9).

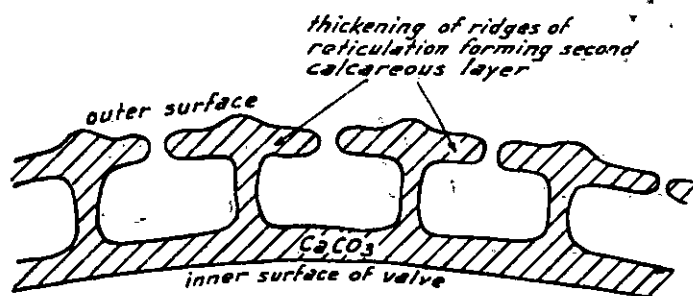


Fig. 9: Schematic section through a spongeous type of ornamentation.

Taxonomic importance of the ornamentation

In order to explain the importance of ornamentation for Ostracod taxonomy, it is necessary to distinguish between "primary ornamentation" and additional or "secondary ornamentation" (fig. 10). Primary ornamentation includes the presence, shape and approximate position of the main ornamental structures on the lateral outer surface of the valves, i.e. the more prominent ridges, costae, lobes, etc. Also included in this term is the basic type of surface ornament e.g. pitted, tuberculate, spinose, spongy, or absence of any ornamentation, viz. smooth.

"Secondary ornamentation" embraces the additional ornamental details over or in between the primary ornamental elements such as, for example, intercostal reticulation, presence or absence of posteroventral spines (see p. 10), etc. The degree of development and the detailed appearance of the primary ornamental elements may also be considered as ornamental features of secondary importance.

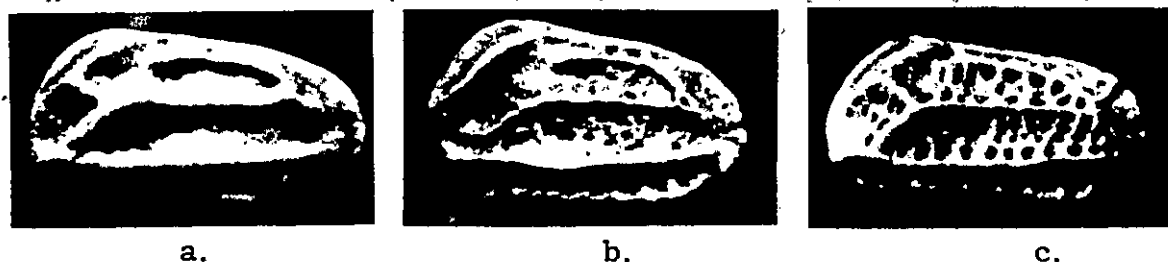
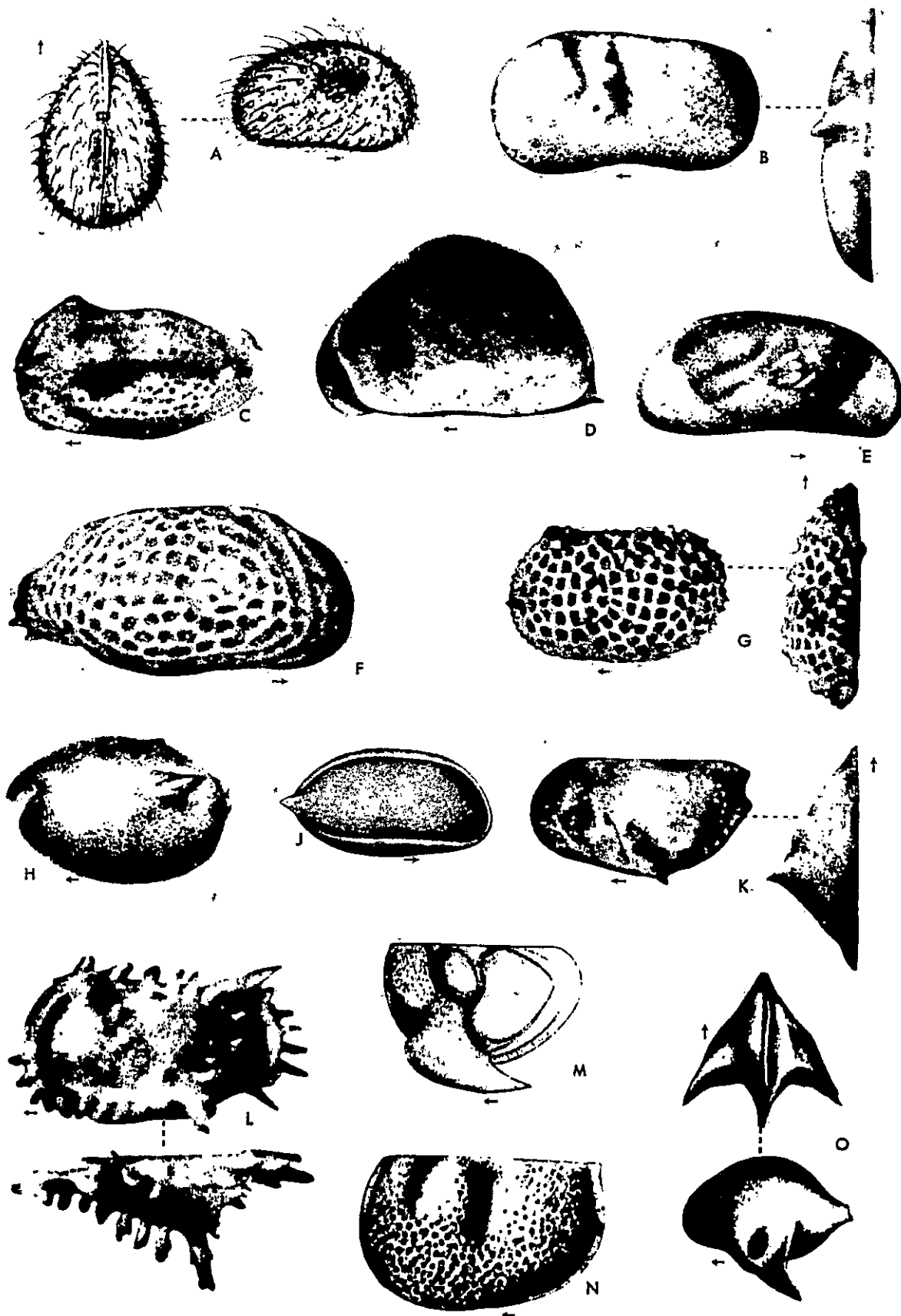


Fig. 10: Three species of the Middle Jurassic genus Pleurocythere illustrating the difference between primary and secondary ornamentation.

- a. Pleurocythere richteri, without secondary ornamentation, showing the primary ornamentation for the genus (i. e. the basic arrangement of the main ridges).
- b. Pleurocythere regularis and c. Pleurocythere connexa, showing secondary ornamentation (reticulation) between the main ornamental ridges. (After TRIEBEL, 1951)



5. Fig. 11 : ORIENTATION OF THE OSTRACOD CARAPACE
(mainly after TRIEBEL, 1941)

Next to shape, ornamentation is the most important taxonomic feature in Ostracod valves from generic level downwards, and its importance for generic and specific determinations is strongly emphasized. As a rule, species belonging to one genus show a remarkable constancy of primary ornamentation. Although a similar type of primary ornamentation (as defined above) may of course occur in different genera, it is exceptional for two species of the same genus to differ in primary ornamentation.

All species of Macrocypris, for example, are smooth; Cyprideis species are generally punctate; Trachyleberis is always tuberculate; Protocythere species all have the three prominent longitudinal ridges; Pleurocythere shows the typical arrangement of costae as shown in fig. 10, etc., etc. Shape or outline, combined with primary ornamentation in many cases are sufficiently characteristic to recognize a genus without taking into account additional details such as hinge, pore canals, muscle scar, etc. This is especially helpful in the frequent cases, when only closed carapaces of fossil forms are available and the inner structures of the valves cannot be studied.

Within a species, the ornamentation is as a rule remarkably constant, even down to the smaller details of secondary ornamental features. For distinguishing different species of one genus, the ornamentation is one of the most helpful characteristics in the valves. Slight differences are often found to be of a constant nature, and these may then be used as characteristic features for distinguishing species or subspecies. In well sampled profiles it may often be observed that certain sculptured forms often rather quickly pass through a number of minor, but nevertheless clearly noticeable changes of ornamental details. At each level, however, such changes are constant and may be traced laterally over wide areas, while their succession often proves to be identical in different areas. Such phenomena are to be considered as gradual morphogenetic changes, which, independent of environment, are due to the advance of geologic time. As such, they are ideally suited for local or regional isochronous correlations.

In the literature a few of such cases are recorded (e.g. for Lower Cretaceous Cytherelloidea (illustrated in fig. 12) and Protocythere by BETTENSTAEDT, 1958 and a Bathonian species of Lophocythere by LUTZE, 1958). A careful study of ornamented species in well-sampled profiles will undoubtedly reveal the existence of many more similar gradual morphogenetic changes in Ostracods. Their usefulness for time-stratigraphic synchronous correlations need not be stressed.

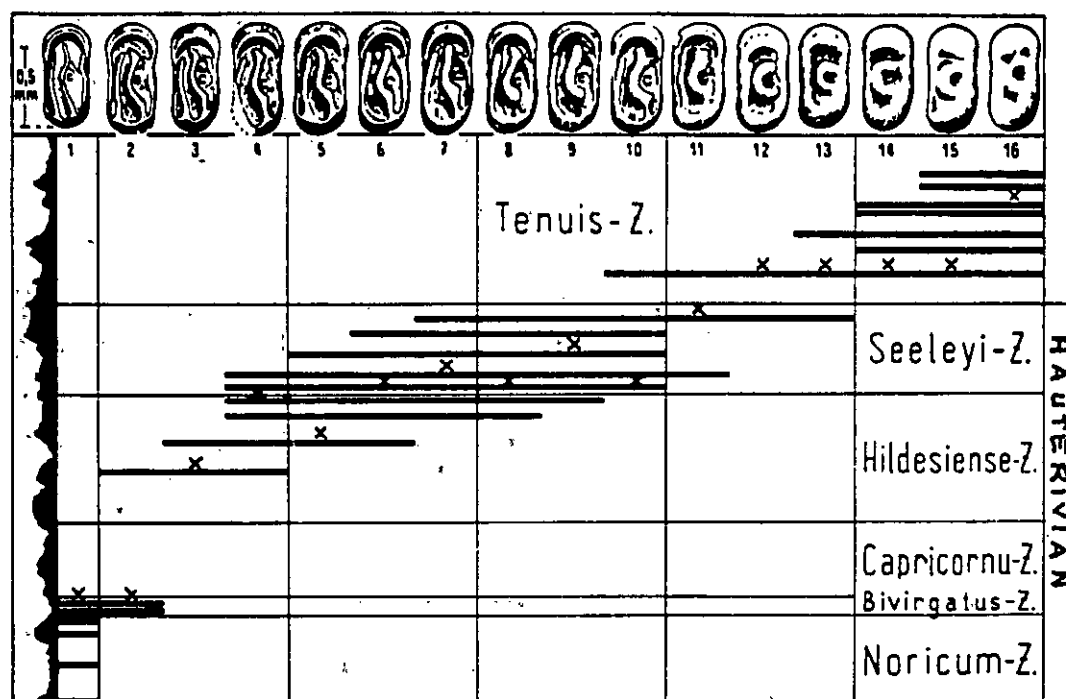


Fig. 12: Showing gradual change of ornamentation in Cytherelloidea ovata WEBER s. l. through the Hauterivian of N. W. Germany. Horizontal lines indicate the limits of variation in 22 populations, arranged stratigraphically as parts of the complete morphogenetic series shown at the top. Crosses indicate the stratum typicum for each of the figured specimens (fig. 1-16). (after BETTENSTAEDT, 1958)

Relation between ornamentation and mode of life

There seems to be a definite relationship between the degree of ornamentation of Ostracod valves, and the habitat in which the animals live. This relationship may be shortly summarized as follows:

In freshwater Ostracods the valves are practically all smooth or weakly punctate. The pronounced sulci and/or the presence of hollow subcentral tubercles as found in certain freshwater genera, such as Ilyocypris and Limnocythere, give such forms a rather strongly ornamented appearance, but these features are not, strictly speaking, part of a true ornamentation as defined above.

Brackish water genera are not usually perfectly smooth. They are either punctate (Cytheromorpha, Leptocythere, most Cytherideinae) or occasionally provided with protuberances (some spp. of Cypridea) or ridges (Macrodentina).

Strongly ornamented forms live exclusively in marine habitats. The large majority of those belong to the family Cytheridae of the Podocopa.

The heaviest type of ornamentation occurs in marine forms, especially among the species living in littoral and shallow neritic environments (Carinocythereis, Hemicytherura, Bradleya, Mutilus, Callistocythere, etc., etc.). Smooth forms also occur in these habitats (noticeably those living on plants, such as the Paradoxostominae, and some marine Cyprids) but with increasing depth the percentage of ornamented Ostracods in marine assemblages diminishes clearly, more and more smooth forms appear (Krithe, Argilloecia, Cytherella, Bythocypris), and in the bathyal sphere only a few ornamented species of predominantly neritic genera such as Trachyleberis, Echinocythereis, etc. still occur.

It may generally be concluded that ornamentation, which obviously increases the strength of the valves, is found mainly in forms that live in more or less turbulent environments (littoral, shallow neritic) where deposition of coarse sediments and wave action prevail.

The amount of calcium in the water seems to exert a certain influence on the thickness of the calcareous valves, and probably also on the strength of development of the ornamental structures, but in practical work such cases hardly ever become apparent.

6. Pores

The valves of Ostracoda are pierced by pores which on the exterior surface of the shell bear hair-like bristles.* These bristles are important sense-organs, and they are connected with the central nervous system by nerves running through these pores. In the calcareous valves two different categories may be distinguished: normal (or lateral) pores and marginal pore canals.

Normal or lateral pores

These are holes which run more or less perpendicularly through the outer lamellae. They are found scattered over the lateral surface of the calcareous outer lamellae. The marginal zone, where outer and inner lamella are fused, is not pierced by normal pores, since these would run from outer world to outer world, and not from outer world to the inside of the Ostracod body.

The number of lateral pores is very variable, ranging from several hundreds per valve to almost none. Their distribution often seems to follow a definite pattern. Many Ostracods have so many pores that it is difficult to distinguish any such pattern at all, but in species with less numerous lateral pores it can often easily be observed that their number and distribution (apart from being nearly the same in right and left valve) is also similar in all specimens of one species (see Fig. 13). It is possible that this feature may be of help in distinguishing different species of otherwise practically featureless genera, such as, for example, Cytherella.

* Some pores may also serve as openings for glands situated inside the duplicatures.



Fig. 13 Outside view of two right valves of Paracyprideis rarefistulosa (LIENENKLAUS), showing the similarity in the distribution of the (sieve-type) lateral pores over the shell surface. (after TRIEBEL, 1941)

Two different types of lateral pores may be distinguished.

- a) Open (or single) lateral pores (Fig. 14a).
These are just holes through the calcareous valve; they are rounded or subrounded in outline and generally rather small. In some forms they widen towards the interior.
- b) Sieve-type lateral pores (Fig. 14b, c).
These pores, towards the outer surface of the shell, are surrounded by a number of small blind pores, giving the impression of a sieve. MÜLLER (1894) has suggested their possible function being that of light-sensitive organs.

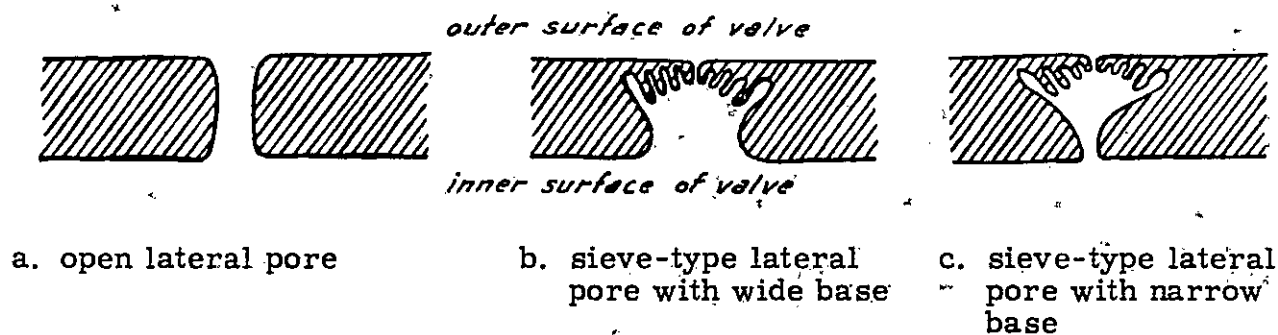
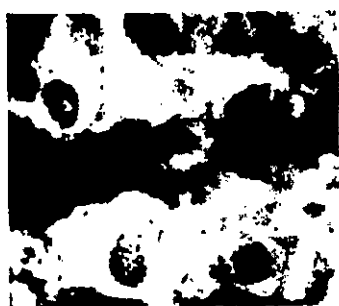


Fig. 14 : Schematic section through different types of lateral pores.

Sieve-type pores are larger than open pores, and it is likely that all lateral pores of more than medium size are sieve-type ones. They are normally rounded or subrounded (Fig. 15a, b), but in certain cases they become slit-like towards the outer surface as, for example, in certain species of Cyprideis and Loxoconcha (Fig. 15c).



A. Cnestocythere
(Miocene)



B. Paracyprideis
(Oligocene)



C. Cyprideis
(Recent)

Fig. 15: Sieve-type pores as seen on the outer surface of the valves.
(after TRIEBEL, 1941, 1950, 1956)

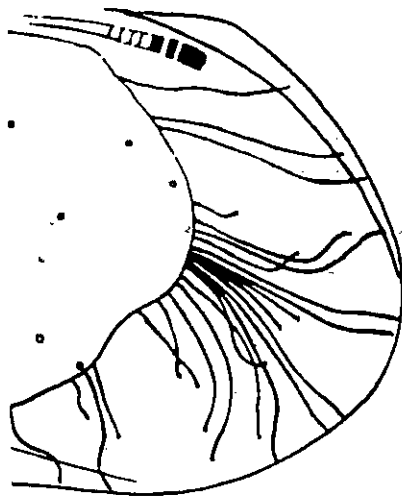
The type of normal pores is generically constant, and it would appear that at higher level, too, they constitute a useful taxonomic feature. Sieve-type pores are so far known to occur only in the family Cytheridae, and within this group they are probably restricted to certain subfamilies only. So far they have been reported mainly from the Cytherinae, Hemicytherinae (always with wide base, see Fig. 14b), Progonocytherinae, Cytherideinae (often with narrow base, see Fig. 14c), Loxoconchinae and Xestoleberidinae. They probably do not occur in the important subfamily Trachyleberidinae.

Marginal pore canals (Fig. 16)

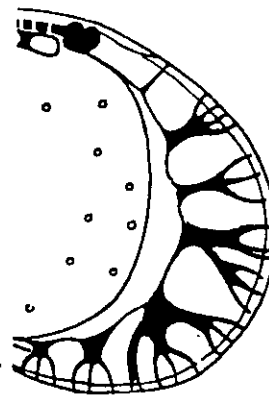
These are long, drawn-out pores situated in the marginal zone. They always originate at the line of concrescence and run in the plane of fusion of inner and outer lamella either to the outer margin (true marginal pore canals, see Fig. 16d), or they reach the outer surface of the lamellae somewhere between the line of concrescence and outer margin, in which case they are termed false marginal pore canals (Fig. 16a, e). False marginal pore canals running partly through the inner lamella, always reach the outer surface distally of the selvage; in other words, they never surface on the proximal part of the inner lamella between selvage and line of concrescence. In many forms the marginal pore canals, before reaching the outer margin, divide themselves into two or more branches. Such pore canals are known as "branching" or "bifurcating" ones (Fig. 16b), whilst non-branching ones are termed simple pore canals.* Marginal pore canals often show a bulbous enlargement near the middle of their length (or somewhat more towards the outer margin) marking the base of the hair or bristle they carry at their distal end (Fig. 16e).

Marginal pore canals are an important taxonomic feature of Ostracod valves. Their number varies greatly in the different species and genera. In the anterior marginal zone, where the marginal pore canals are practically always more numerous than in the ventral and posterior parts of the marginal zone, their number may vary between 5 and about 100 (Fig. 16c, d).

* Separate simple pore canals, in lateral view, sometimes cross (or even partly cover) each other. These may erroneously be considered as true branching pore canals (see Fig. 16a).



a. Simple, curved anterior marginal pore canals (incl. many false ones) in Semicytherura.



b. Branching anterior marginal pore canals in Callistocythere.



c. Numerous, close-set, curved anterior marginal pore canals in Cytheretta.



d. Few, straight, widely spaced anterior marginal pore canals in Palmenella.



e. Few, simple, curved anterior pore canals (incl. some false ones) in Pleurocythere.



f. Posterior marginal pore canals in Pleurocythere.

Fig. 16: Different types of marginal pore canals in the Cytheridae (Podocopa).

The approximate number of marginal pore canals, as well as position and general shape seem to be basically the same in species belonging to one genus. Within a species, all individuals show only very minor differences in number, form and relative position. For example, the number of anterior marginal pore canals in all species of Loxoconcha seems to lie between 7 and 10, all being simple, straight and widely spaced. In Loxoconcha elliptica, this number seems to be constantly 9, and specimens with 8 or 10 are exceptionally rare. Similar cases are plentiful; they may be most easily observed, of course, in species with relatively few marginal pore canals.

Conspicuous, widely branching marginal pore canals are present only in the family Cytheridae of the Podocopa, and a few Cyprids, like Paracypris.

Pseudo-marginal pore canals are found in certain hemicytherine genera as Caudites and Orionina. These are a result of a peculiar secondary fusion between inner and outer lamella.

7. Hinge

The two valves of an Ostracod are joined at their dorsal regions. Normally some structure is present to keep the valves attached dorsally when the animal opens its shell for the various processes of life. In the Myodocopa and Cladocopa the valves are connected by an elastic ligament and no special hinge structure is present in the calcareous parts.* In the Cytherellidae this ligament has not yet been observed and no calcareous hinge elements are present either. In the other Platycopa and in all Podocopa, however, the calcareous parts in the dorsal marginal region are more or less differentiated into grooves or sockets and ridges or teeth in the zone of attachment. These elements are modifications of the inner lamellae (mainly the selvage) and possibly also of the outer lamellae; together they are generally known as "the hinge".

The hinge in Ostracods plays an important part in their taxonomy. Unfortunately, the terms used for the different types of hinge, and those for describing the various hinge elements, have not always quite the same meaning in the literature. Some have been borrowed from those used to describe the lamellibranch hinge. These are not altogether appropriate in their new context and may lead to ambiguity. In order to avoid vagueness of meaning, a set of terms which seem adequate and unequivocal for the purpose of referring to hinge elements or hinge types has been adopted for this manual. These are presented and defined below.

a. Terminology of hinge elements

The hinge in Ostracods may be either simple, i.e. consisting of one element in each valve, or compound, that is divided into a number of elements in each valve.

* An exception: Conchoecia curta (see SKOGSBERG, 1920)

The simple hinge is one of the three basic types of hinge, viz. the adont type (see p. 26). It is made up of a single element in each valve. This consists of a groove along the dorsal margin of the larger valve (in the Podocopa this is more often the left valve) into which fits the edge of the smaller valve (Fig. 17a, 18A).

The compound hinge consists of three basic elements in each valve, i.e. the anterior, median and posterior hinge elements (Fig. 17b). The terminal elements (i.e. the anterior and posterior ones) are projections beyond the hinge margin (in dorsal view) in the smaller valve which in the majority of cases is the right valve. The terminal elements of the larger (usually the left) valve consist of an anterior and posterior socket, which serve to receive the projecting terminal elements of the smaller valve. The rule that the projecting terminal elements are always in the smaller valve would seem to be without exceptions.

The median hinge element as a rule alternates with the terminal elements, so that if the latter are projecting teeth or ridges, the median element separating them will be a groove, or if the anterior and posterior elements are sockets or grooves, the median element will be a projecting ridge. However, exceptions are known (e.g. Haplocytheridea), where all elements in the larger valve are grooves, while all elements in the smaller one are projecting beyond the hinge margin. But in such cases, too, the constituent elements remain clearly distinct: the terminal elements being more coarsely dentate and/or developed more strongly than the median one.

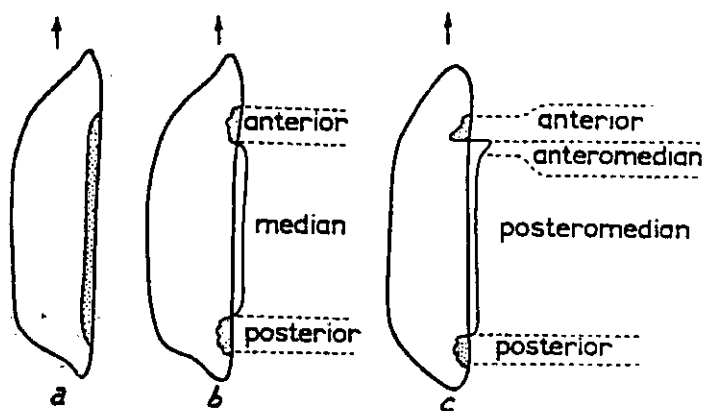
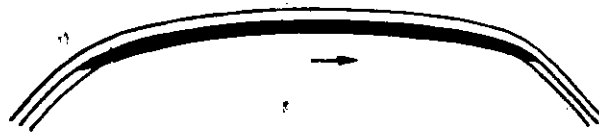
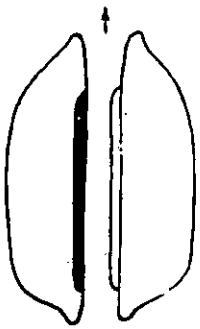
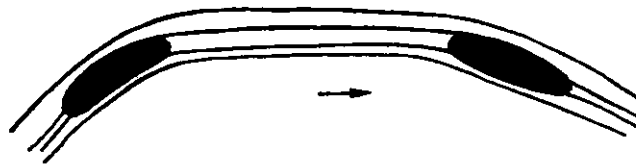
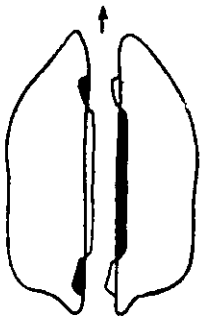


Fig. 17 Dorsal view of left valves of three Ostracods, illustrating division of hinge into "elements". Stippled areas represent sockets. (after SYLVESTER-BRADLEY, 1956)

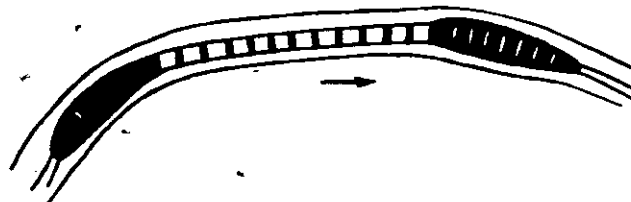
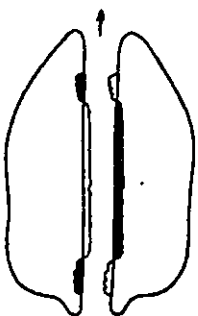
A. THE ADONT HINGE



B. THE MERODONT HINGE



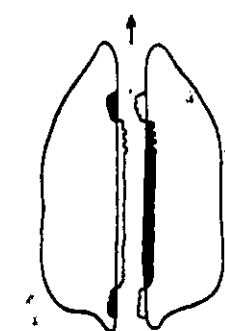
a. Lophodont



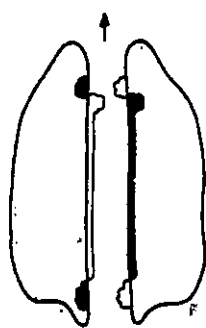
b. Entomodont

Fig. 18: Schematic representation of main Ostracod hinge types. Dorsal view (left) and inside view of left valve (right). The left valve in these figures has been taken to be the larger one. Black areas represent sockets or grooves. (continued on next page)

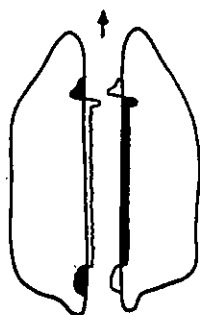
C. THE AMPHIDONT HINGE



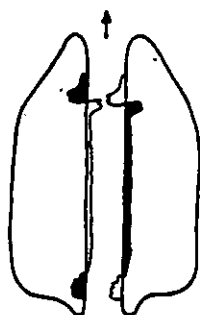
a. Archidont



b. Lobodont



c. Heterodont



d. Schizodont

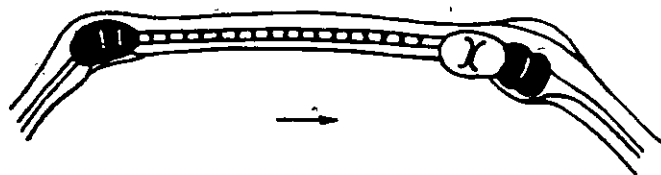
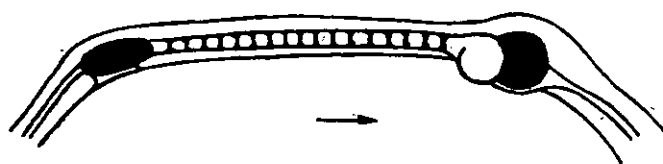
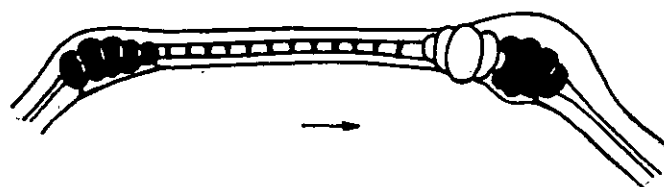
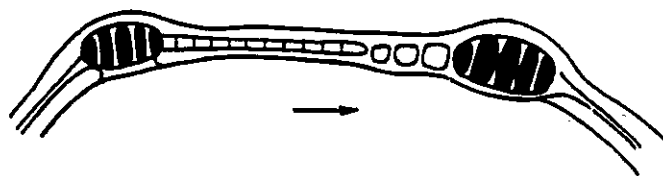


Fig. 16: (continued)

As will be explained below, the compound hinge with three hinge elements evolves into a more advanced type of compound hinge with four elements. This is brought about by a subdivision of the median hinge element into two parts: an anteromedian and a posteromedian element (Fig. 17c). The distinction between these two elements is at first not very pronounced, the anterior one being merely more coarsely dentate than the posterior (Fig. 18C,a). In later, more evolved forms this distinction becomes clearer, the anteromedian element becomes rather similar in proportions to the anterior one, whilst the posteromedian element usually remains much the longest element of the hinge.

III. ECOLOGY AND PALAEO-ECOLOGY OF POST-PALAEOZOIC OSTRACODA

The post-Palaeozoic Ostracoda constitute an admirably suitable group of micro-organisms for ecological and palaeoecological studies for the following reasons:

- a) Recent representatives are known to inhabit practically all types of aquatic environments.* They occur in nearly all freshwater environments ranging from the deepest lakes to the shallowest, very temporary road-side pools; some are typical brackish water forms, and others may be encountered in every type of marine environment, from tidal pools down to abyssal depths below 3000 m. Many species show a great tolerance for fluctuations of environmental conditions like temperature and salinity, and are able to flourish in such extreme environments as hot-water springs, highly saline pools, etc. Numerous other groups, however, are restricted to well-defined biotopes and susceptible to slight changes of one or more commonly used environment- or biotope-determining factors. They generally occur in large numbers, and because of their minute size even a small sample may contain numerous representatives of living or fossil forms.
- b) Apart from most Myodocopa, many of which are pelagic, thin-shelled, and hardly encountered in the fossil state, the majority of Ostracods possess a hard calcareous carapace which may fossilize relatively easily. The larvae of these benthonic Ostracods would seem to have no pelagic stage, and the ontogenetic development is restricted to one biotope (see ELOFSON, 1941).
- c) In many cases there is a distinct relationship between genotypic structural details and auto-biotope, and certain phaenotypic structures dependent on environmental changes within the auto-biotope are also known.
- d) Conspicuously different taxa inhabit the more widely separated, largely differing auto-biotopes, and as a result thereof their auto-thanatotopes, too, are often confined to reasonable limits and do not normally overlap.

* A few rare instances of Ostracods inhabiting terrestrial environments have also been reported (KLIE, 1925; MENZEL, 1923; HARDING, 1953), but these are not further considered here.

e) The study of the palaeo-ecology of many post-Palaeozoic forms is greatly facilitated by the fact that of most genera (especially Tertiary ones) one or more species are still living at the present time.

Relationships of Ostracods to the main environmental factors

In the following remarks the influences on Ostracods of the four important environmental factors: salinity, temperature, substratum and depth are briefly discussed.

a. Salinity

ELOFSON (1941) and WAGNER (1957) have shown that the behavior of Ostracods with respect to the salinity of the water complies with the observations of REMANE (1934) on other groups of organisms, namely that the number of marine species decreases regularly down to a salinity of abt. 10°/oo, but shows an abrupt drop in water of less salinity, while the number of freshwater species diminishes very quickly when the salinity surpasses 3°/oo. In the meio-mesohaline range (3-10°/oo) Ostracod species are not numerous, but they often occur in large numbers. They may be said to form the true brackish-water species, and since the salinity in brackish environments may still vary considerably, they may therefore be regarded as more or less euryhaline forms.

Cyprideis torosa (JONES), for example, flourishes in N.W. European waters with a salinity ranging from 2 to 17°/oo. The species Cyprideis edentata KLIE; however, seems to inhabit much more saline environments, occurring on Bonaire in pools of 26-80°/oo salinity. This example clearly shows that congeneric species often have considerably different autotopes, and it also illustrates the danger of generalizing ecological data of only one or two species to generic level.

Apart from true brackish-water species, the mesohaline environments may be inhabited by certain species belonging to otherwise predominantly freshwater genera (like, for example, some species of Candona), and also by others belonging to mainly marine genera* (some species of Loxoconcha, Leptocythere, Xestoleberis, etc.). The total number of species inhabiting typical brackish-water environments, however, remains very small compared to the host of Ostracod species living in true freshwater or marine environments.

Salinity-indicating structural details in Ostracods

Freshwater forms (fresh-oligohaline) - Mainly smooth, thin-shelled forms, practically all belonging to the Cyprididae (muscle scars!). Hinge type predominantly adont; amphidont hinge never present; merodont hinge very rare. Caudal process, hinge ears, and conspicuous branching marginal pore-canal absent. Normal sieve-type pores extremely rare (Limnocythere). Eye spots never strongly developed.

* Mostly genera living in the shallower part of the neritic zone (mainly the littoral zone), the majority of which are euryhaline.

Brackish-water forms (mesohaline) - Usually rather thick-shelled, ornamentation weak, often punctate. The majority of species belongs to the Cytheridae (mainly to the Cytherideinae, some to the Cytherurinae and Loxoconchinae). Hinge merodont or modified amphidont; rarely strongly heterodont. A number of species, mainly of genera belonging to the Cytherideinae (such as Cyprideis, Cytheridea, Cytherissa, etc.) develop typical phaenotypic hollow tubercles on the lateral surface of their valves when moving within their brackish water biotope to environments of less salinity. Their presence or absence, also in fossil forms like Haplocytheridea and Neocyprideis, may be taken to reflect salinity changes within the mesohaline salinity range in the environments inhabited by these forms during life, i.e. environments included in their biotope. The tubercles found on some species of the well-known Upper Jurassic/Lower Cretaceous freshwater genus Cypridea are of a different type: they are presumably genotypic ornamental structures (not hollow, rather constant in number and position within a species) and thus probably not indicative of salinity changes. The hollow tubercles forming on certain freshwater forms in calcium-deficient water likewise do not indicate a salinity change.

Marine forms - Practically all species exhibiting a rostrum, a strong amphidont hinge, conspicuous branching marginal pores, heavy ornamentation, well-developed eye spots, sieve-type normal pores, and/or a complete fusion of inner and outer lamella live in marine environments. However, other species not showing these structures also occur in marine waters; often so in appreciable quantities.

The ecological significance of the above-mentioned structural details has been gathered from our present overall knowledge of recent Ostracods and their auto- and syn-biotopes. There is no reason or contradictory evidence so far, which would suggest this relation between salinity and morphological structures of Ostracod valves to have been considerably different in the past from what it would appear to be at present. For an evaluation of the salinity factor in past environments, Ostracods may thus be of considerable help, provided the necessary attention is paid to determine whether the fossils were deposited inside the area of their biotope or not. At the recent level freshwater Ostracods, after death, are often rather easily transported by rivers, etc., to brackish and shallow marine environments, and the thanatotopes of fresh- and brackish-water Ostracods often considerably overlap the biotopes of marine (mainly shallow marine) associations. In the past, too, this must have happened quite regularly, and in every attempt to determine the "palaeo-salinity" (as OERTLI, 1957, terms it) of a fossil environment, this possibility should first be seriously considered.

b. Temperature

The tolerance of heat and cold differs in Ostracod species reproducing in the early spring and those reproducing during the summer. The tolerance of cold for certain species which reproduce in cold temperature is very great. They even withstand being frozen fast in ice for days without apparent ill effects. At warm temperatures, however, they quickly expire. Tropical species, and forms which reproduce during the warm summer months in the temperate zones, die in cold water but maintain normal living processes at very warm temperatures.

Some Ostracod species are known to prefer warm water, and others are occasionally found to flourish in what might be termed hot water. KLIE (1939), for example, describes a Heterocypris species from a hot-water spring in Algiers with an optimum temperature of 40-46° C (minimum 35°, maximum 51.5°). A number of Ostracod species would seem to be restricted to the cold water regions either in the north or south high latitudes, or at depths which have cold water all year round. ELOFSON (1941) found different groups of species to be more or less restricted to arctic, subarctic, boreal, mediterranean, etc. areas. He reports that representatives of some arctic-boreal species occur at greater depths in boreal areas than in the arctic region, presumably so avoiding the higher summer temperatures of the surface water in boreal areas. All such forms are more or less stenotherm, and the temperature is one of the major factors determining their biotopes. WAGNER (1957) has drawn attention to the practical usefulness of such stenotherm Ostracods for deducing the palaeo-climatological conditions during the Pleistocene as caused by the successive glacial periods.

As far as is known at present, the only structural feature of Ostracod valves which would seem to be affected by differences in temperature is their size. Arctic representatives of certain arctic-boreal species sometimes attain as much as twice the size as specimens living in the more boreal regions.

In general, the number of marine Ostracod species increases towards the tropical belt, where it reaches its maximum, and a considerable number of genera do not occur outside typical tropical environments. Since their fossil representatives presumably behaved likewise, they may be regarded as indicative of past tropical conditions, too.

As regards the relation of warmth to the rate of growth in Ostracods, it has been demonstrated that increase of temperature accelerates the growth rate. In certain cases the time required in nature for the development from egg to sexual maturity may be reduced to 50%, when keeping cultures at constant relatively warm temperature.

It is clear that the relation of temperature to the Ostracod in nature is a complex ecological problem. The amount of heat supplied to an environment influences numerous other factors, such as the food supply, thermal gradient of the water, the nature of the bottom sediment (presence of coral fragments, etc.), the vegetation, the hydrogen ion concentration, etc. Therefore, the increase of a species is much more complicated than the growth of a specimen in its relation to heat. Field observations show each species to reach its greatest abundance when the temperature of the water is most favorable for its reproduction and growth, but at present specific data are insufficient to determine whether the temperature is the prime factor responsible for this or to what extent other factors pertaining to habitat and biotic relationships play an important role.

c. Depth

Freshwater

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Most freshwater Ostracods live in shallow aquatic environments, and only few species would seem to occur predominantly in the deep parts of large

lakes, etc. Among the latter are quite a few species of Cytherissa and some typically trapezoid-shaped species of Candona, often referred to the subgenus Lineocypris.

A number of Ostracod species have been reported to live typically in ground water or other subterranean environments. These include certain species of Candona, Cypridopsis and Sphaeromicola. Others inhabit springs (some species of Ilyocypris, Ilyodromus, Potamocypris, Eucypris, etc.) or other flowing waters, but the majority of freshwater forms live in less turbulent or stagnant waters. Some burrow in the bottom sediments, others live on the bottom surface, while quite a few species are equipped with natatory setae on their antennae allowing them to swim about freely. True pelagic freshwater species, however, have so far not yet been reported. Special mention must be made of Notodromas which clings upside down to the water surface.

It is difficult to present general ecological information about freshwater genera, since congeneric species may be found in quite different environments. A typical example presents the well-known genus Candona, which embraces species living in deep-water lakes, shallow pools, subterranean waters, and even one or two oligohaline to meiomesohaline forms.

Brackish water

Brackish water environments are as a rule rather shallow and very little can be said about the depth distribution of species confined to these habitats.

Marine water

Recent Ostracods are known from practically every type of marine environment ranging from very shallow tidal pools to depths exceeding 3000 m. The large majority of the marine species, however, is found in the littoral and epi-neritic zones.

Most published data on the bathymetric distribution of recent marine Ostracod species and genera are based on the occurrences of their empty valves only, and therefore these cannot be taken as very reliable indications of their actual biotopes. The detailed study of ELOFSON (1941) shows that a number of marine species in the Skagerrak would seem to live within rather well-defined bathymetric limits, but the total of our knowledge about the relationship between living Ostracods and depth is still very scanty, and does not allow any too detailed general conclusions to be drawn. Nevertheless, by including the wealth of information regarding the bathymetric distribution of empty valves and carapaces, it is possible to construct an approximate picture about the thanatotopes of marine Ostracods at different depths, which is still useful for a further ecological differentiation of marine faunas into three types of different bathymetric range, i.e. littoral, shallow marine (epi- and ? infra-neritic) and deep marine (bathyal, ? abyssal) faunas. Each of these types of marine faunas is characterized by a number of genera more or less restricted to it. A number of other genera, however, are not restricted to one type of bathymetric environment, and different species of these may be encountered in marine environments of very variable depth. They include such common genera as, for example, Bairdia, Cytherella and Cytheropteron.

The bulk of marine Ostracod species and genera lives in the littoral and shallow marine zones, some exclusively in the littoral or the shallow marine zone, and species of others in both.

In palaeo-ecological studies the differentiation between littoral and shallow marine Ostracod assemblages is rather difficult, since their syn-thanatotores necessarily overlap somewhat. Their area of overlap tends to fall within the shallow marine zone rather than in the littoral one. There are, moreover, no conspicuous structural details by which littoral forms may be distinguished from epi- or infra-neritic groups of species. Both groups, however, comprise a number of genera exhibiting certain structural features of the calcareous valves which are rather uncommon both in deeper marine and in fresh to brackish water assemblages. Since the syn-thanatotores of littoral to shallow marine Ostracod assemblages as a rule practically do not overlap those of freshwater and deeper marine ones, these structures normally greatly facilitate a recognition of the littoral to shallow marine environments in which such Ostracod assemblages were deposited. These structural features are the following:

Heavy ornamentation
Strong amphidont hinge
Conspicuously branching marginal pore canals
Well-developed eye spots

Consequently, deeper marine Ostracod assemblages, as already mentioned by HEJJAS (1894), comprise mainly thin-shelled, translucent, mostly non-ornamented species, seldom provided with well-developed hinge teeth. As such they consist mainly of species belonging to the genera

Cytherella
Bairdia
Bythocypris
Macrocypris
Argilloecia
Krithe
Pseudocythere
Bythocythere
Polycope
Cytheropteron) (smooth species)

Often such faunas also include some representatives of more ornamented genera like Henryhowella, Trachyleberis, Echinocythereis and/or Bythoceratina. Together they form a very typical type of Ostracod assemblage, which is found in nearly all the deeper parts of the present oceans. It is also regularly encountered in the fossil state; especially in some Upper Cretaceous strata of NW Europe. Cytherella and Bairdia, both exclusively marine and usually present in considerable quantities, are easily recognized and immediately distinguish these deeper marine assemblages from the likewise thin-shelled, non-ornamented, weak-hinged freshwater assemblages, consisting of mainly Cyprididae. Usually this type of assemblage in recent as well as in fossil material is accompanied by a foraminiferal fauna showing a high percentage of pelagic species.

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The influence of temperature on the depth at which certain arctic-boreal species occur has been discussed.

For the sake of completeness it must be mentioned that a number of marine Ostracods are pelagic. Since these belong exclusively to the Myodocopa, which have practically no fossil representatives, they are not further considered here. It is interesting to note that the females of some of these Ostracods (e.g. Philomedes) are known to lead a benthonic life, but after reaching maturity they become temporarily pelagic for the purpose of copulating with the short-living pelagic males.

d. Substratum

Reliable data pertaining to the relationship of living Ostracoda to the substratum are very scarce, especially as far as the important group of marine forms is concerned. Again, the most detailed available study of this subject is ELOFSON's (1941) paper dealing with the marine Ostracoda from around Sweden. This study shows the very important role of the substratum as a biotope-determining factor for many Ostracod taxa, and it includes a detailed analysis especially of the influence exerted by the substratum on certain important structural features of Ostracod valves. As stated before (see p. 13), however, it would seem advisable to postpone the application of ELOFSON's findings to the whole order until more additional information becomes available from other areas, especially from the practically unexplored tropical regions with their abundance of Ostracod life.

It is useful, however, to summarize some of ELOFSON's more important observations here, if only to emphasize the fact that a great deal of attention should be paid to the substratum in every ecological study of recent Ostracods, and further to demonstrate that in general a close relationship between the structure of the carapace and the nature of the substratum undoubtedly exists in marine Ostracods.

ELOFSON observed that:

- a) Among species living mainly on plants, a thin-shelled, hyaline, elongate, more or less laterally compressed, weak-hinged and smooth carapace is the most common type (Paradoxostominae).
- b) The same type of shell is often found in the relatively few species which live exclusively in very coarse-grained bottom sediments (shell fragments, gravel).
- c) Species burrowing in the substratum show a twice as high percentage of smooth forms than species living on top of the bottom sediment.
- d) The burrowing forms have as a rule a more strongly calcified shell than the non-burrowing benthonic forms.
- e) Species crawling over the surface of the bottom sediments often exhibit a pronounced flattening of the ventral surface of the carapace.
- f) As regards size, species living on or in sandy bottoms on the whole are shorter than plant-dwelling forms, and the largest forms are found among the species living on a soft muddy bottom.

IV. PREPARATION METHODS

a. Washing

Samples may be washed according to the normal standard method used for Foraminifera, but special attention must be paid to the following points:

i) The carapaces of many Ostracods are often so hermetically sealed that the water cannot replace the air inside, and consequently these carapaces float at the surface in the sieves used for washing. Separate valves of thin-shelled smooth Ostracods often remain at the surface too. Special care must be taken that the water does not overflow the edge of the sieves, since this may cause a near-complete loss of the Ostracod fauna.

ii) The actual washing should be carried out gently. Any hard rubbing of the fractions (especially the coarse - 40 mesh - fraction) over the sieves is to be avoided, and may result in the complete splintering up of valves, which by careful washing might have been recovered undamaged.

b. Picking

Ostracoda on the whole occur in smaller quantities in residues than the Foraminifera, and for obtaining representative Ostracod faunas an extensive picking of the residues is necessary. Experience has shown that in view of their average size, the dried residue is best split in separate fractions with the aid of a set of table sieves of 20, 40, and 80 mesh.

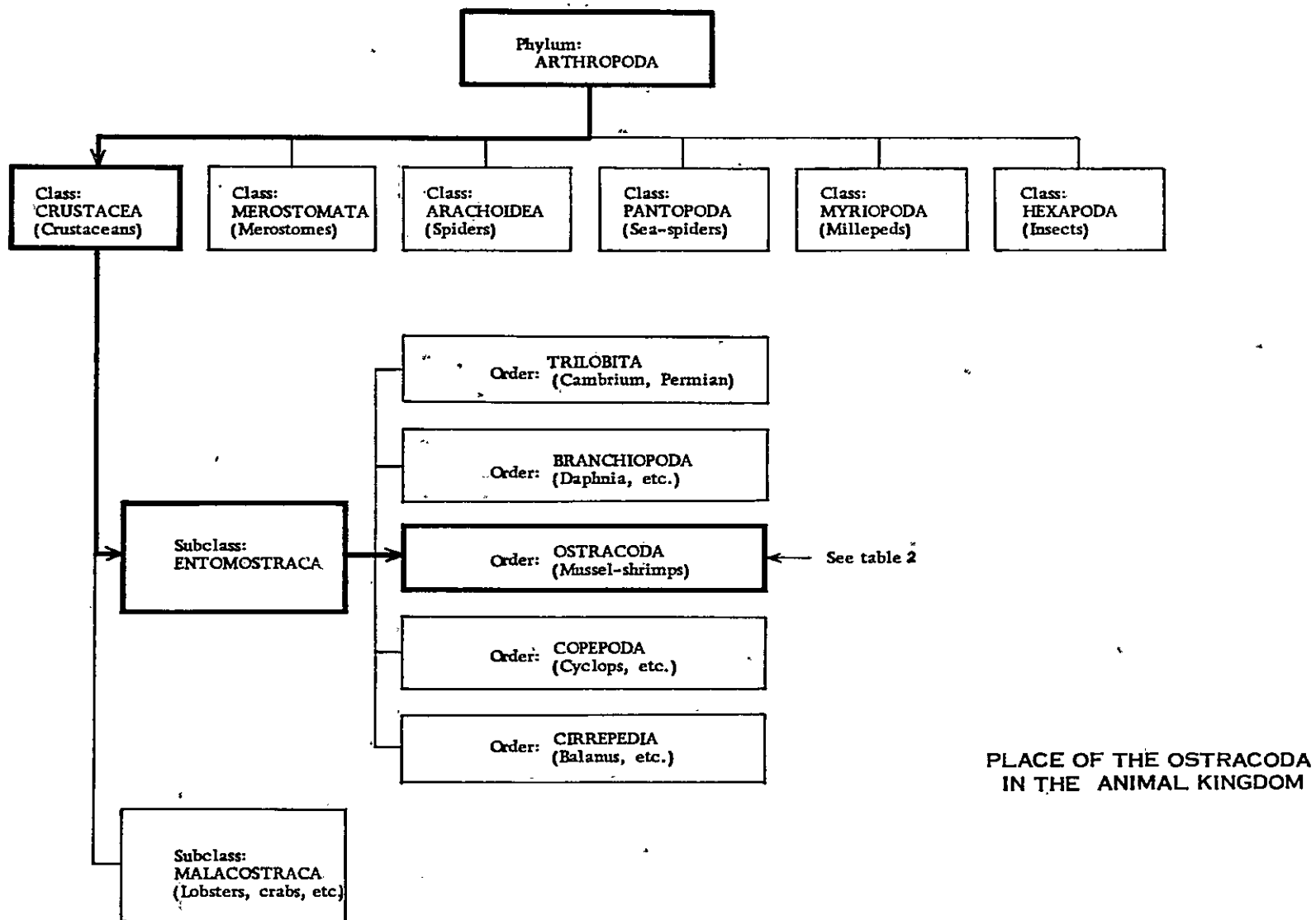
The 20 mesh fraction only rarely contains Ostracods; the bulk remains on the 40 and 80 mesh sieves, and these fractions should be picked as extensively as time permits, and preferably completely. The fraction passing the 80 mesh sieve as a rule contains predominantly immature valves, and not so many adults of small species.

The picking of predominantly sandy residues is considerably speeded up by gently shaking each fraction first in a shallow flat-bottomed receptacle, whereby the organic part of the residue is mechanically sorted and becomes rapidly concentrated in a thin layer on top of the heavier sandy part of the residue. By scraping off and picking this top layer a representative fauna may be obtained in only a fraction of the time required for picking such a sample the normal way.

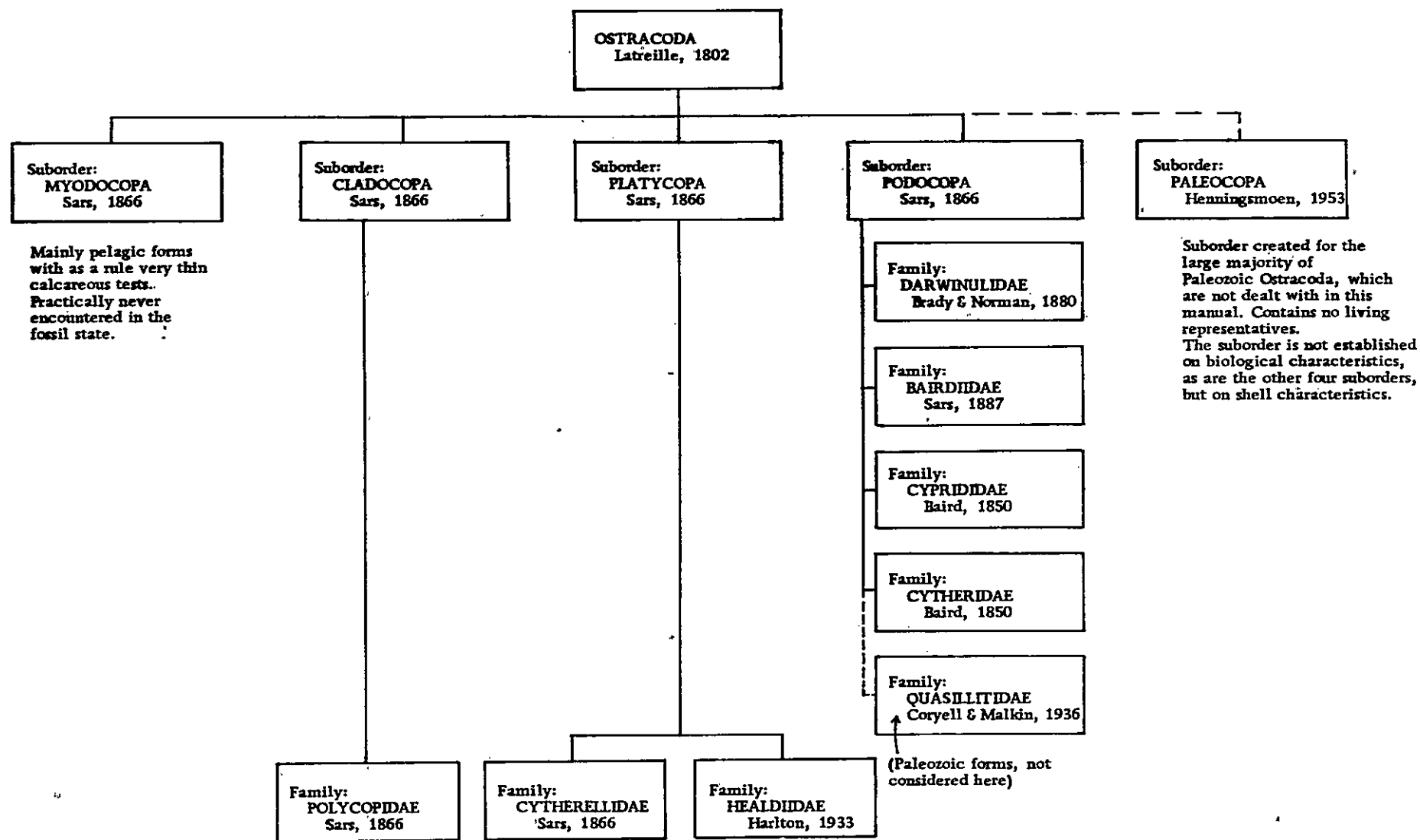
Picking is most conveniently carried out under a binocular microscope using an enlargement of approx. 40 - 50 x.

c. Insoluble residues

The preparation of thin sections of hard unwashable rock samples is not necessary for Ostracod studies, since genera and species cannot be determined in this way. Occasionally such samples contain silicified Ostracods, which may be freed from the calcareous matrix by preparing an insoluble residue of it. As a rule, however, recrystallization has obscured most details of the carapace, rendering the identification of taxa of generic level or below extremely risky, if not wholly impossible. The method is nevertheless sometimes practically useful for obtaining a general picture of the type of Ostracod assemblages embedded in such hard rocks.



V. CLASSIFICATION TABLE I



CLASSIFICATION TABLE 2

Sub-Order	Family	Carapace with an anterior opening	Conspicuous overlap in ventral (oral) region	Peculiar (primitive) calcified inner lamella	Central muscle-scar pattern →	Strong amphidont hinge	Well-developed eye spots	Sieve-type normal pores	Includes strongly ornamented forms	Consistency of the valves
MYODO-COPA	CYPRIDINIDAE	YES	No	No	Normally many scars; arrangement very variable (see p.)	Never	Never	Never	YES	FIRM
	CONCHOECIIDAE	YES	No	No		Never	Never	Never	No	THIN
CLADO-COPA	POLYCOPIDAE	No	No	No		Never	Never	Never	YES	RATHER THIN
PLATY-COPA	CYTHERELLIDAE	No	No	YES		Never	Never	Never	YES	FIRM
	HEALDIIDAE	No	No	YES		Never	Never	Never	No	FIRM
PODO-COPA	DARWINULIDAE	No	YES	No		Never	Never	Never	No	THIN
	BAIRDIIDAE	No	YES	No		Never	Never	Never	YES	FIRM
	CYPRIDIDAE	No	YES	No		Never	Never	Never	Practically none	USUALLY THIN
	CYTHERIDAE	No	YES	No		OFTEN	OFTEN	OFTEN	YES	USUALLY FIRM

Important characters of the calcareous valves in the different suborders and families.